CommerceFlow — Sprint 1A

Documentation Quality Assurance Review v1.1

Status: REVIEW — NOT FINAL

Verified inventory

• OpenAPI: 19 paths; 21 operations; 121 references; 0 unresolved local references.

• 6 editable draw.io files parsed; 12 existing PDFs opened; 5 DOCX files integrity-checked.

• Static artifact checks only; no backend implementation or automated feature tests executed.

Findings to close before FINAL

HIGH — Draft/approved status mismatch

The package contains older DRAFT SRS/ADR/PDF versions alongside approved checkout policy. Cross-document language and status must be reconciled before final release.

HIGH — Idempotency persistence design

The policy defines behavior, but an infrastructure table schema, unique key scope, state transitions, expiry cleanup, and crash recovery require a precise migration design.

HIGH — Security implementation contract

Specify CSRF token bootstrap/rotation, session cookie deployment policy, login filter integration, and consistent 401/403 handling before authentication implementation.

MEDIUM — Checkout snapshot semantics

A retry fingerprint based on a mutable cart snapshot needs a stable persisted initial fingerprint. Otherwise a successful checkout that clears the cart can appear different on retry.

MEDIUM — Cancellation/restock lifecycle

Define exactly-once stock restoration and concurrent cancellation protection, with explicit state transition tests.

MEDIUM — Diagram export fidelity

Companion PDFs are independent previews; they have not been verified as pixel-identical exports of the editable draw.io files.

MEDIUM — API error coverage

Check 400/401/403/404/409/429/500/503 applicability for each operation and document consistent Problem Details responses.

MEDIUM — Test evidence completeness

Traceability identifies planned tests but does not contain executed CI evidence; keep planned and passed statuses distinct.

Release gate

Synchronize source documents and PDFs, specify idempotency storage/recovery, confirm security behavior, visually verify diagrams, then request Tech Lead approval before GitHub PR.